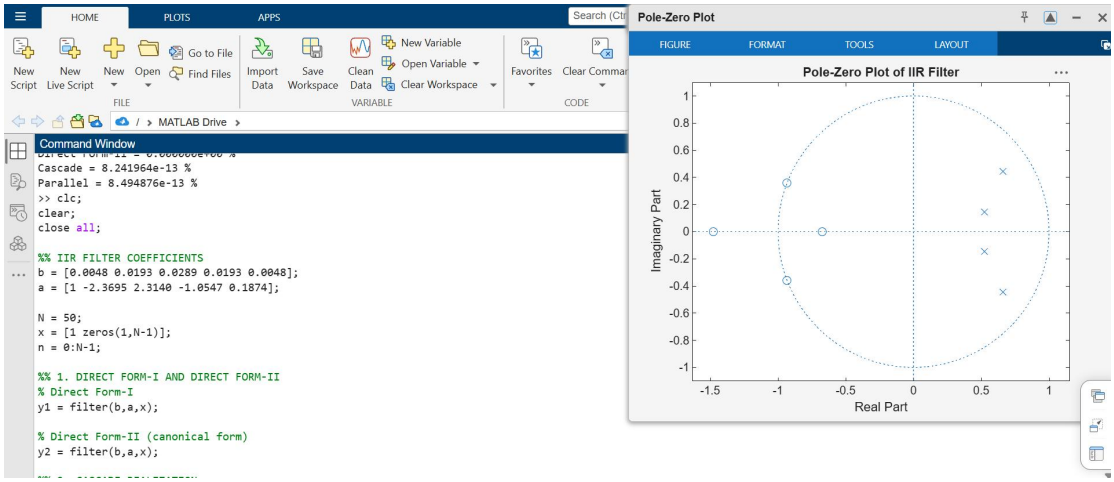
Realization of IIR Filters Using Direct Form-I, Direct Form-II, Cascade and Parallel Structures Using MATLAB

OUTPUT



Command Window

```
=========================================
 IIR FILTER REALIZATION
=========================================

Numerator coefficients b:
    0.0048    0.0193    0.0289    0.0193    0.0048

Denominator coefficients a:
    1.0000   -2.3695    2.3140   -1.0547    0.1874


CASCADE REALIZATION
Second-Order Sections (SOS):
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327

Overall gain:
    0.0048
```